

MACHINE LEARNING | ASSIGNMENT

Supervised Learning on the Telco Customer Churn Dataset

Classification & Regression — comparing two algorithms for each task

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Course **Machine Learning**

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● The Dataset & The Task

ASSIGNMENT BRIEF

- Download a dataset from Kaggle.
- Apply supervised learning for both Classification and Regression.
- Implement & compare at least two algorithms per task.
- Evaluate with appropriate metrics.
- Present comparative results in a clear table.
- Explain every part of the code in the viva.

Telco Customer Churn

A telecom company's customer records — services, contract, billing and whether the customer left.

7,043

Customers (rows)

21

Columns

2

Prediction targets

CLASSIFICATION predict **Churn** (Yes / No)

REGRESSION predict **MonthlyCharges** (a number)

Source: [kaggle.com/datasets/blashtchar/telco-customer-churn](https://www.kaggle.com/datasets/blashtchar/telco-customer-churn)

● Methodology — One Workflow

1 Load

Read the CSV into a table (pandas).

2 Clean

Drop customerID; fix 11 blank TotalCharges (new customers) → 0.

3 Encode

One-hot encoding: turn text columns into 0/1 numbers.

4 Split

80% train (5,634) / 20% test (1,409). Test stays unseen.

5 Scale

StandardScaler puts all features on one range (vital for KNN).

6 Train & Evaluate

Fit 2 models per task; score on the unseen test set.

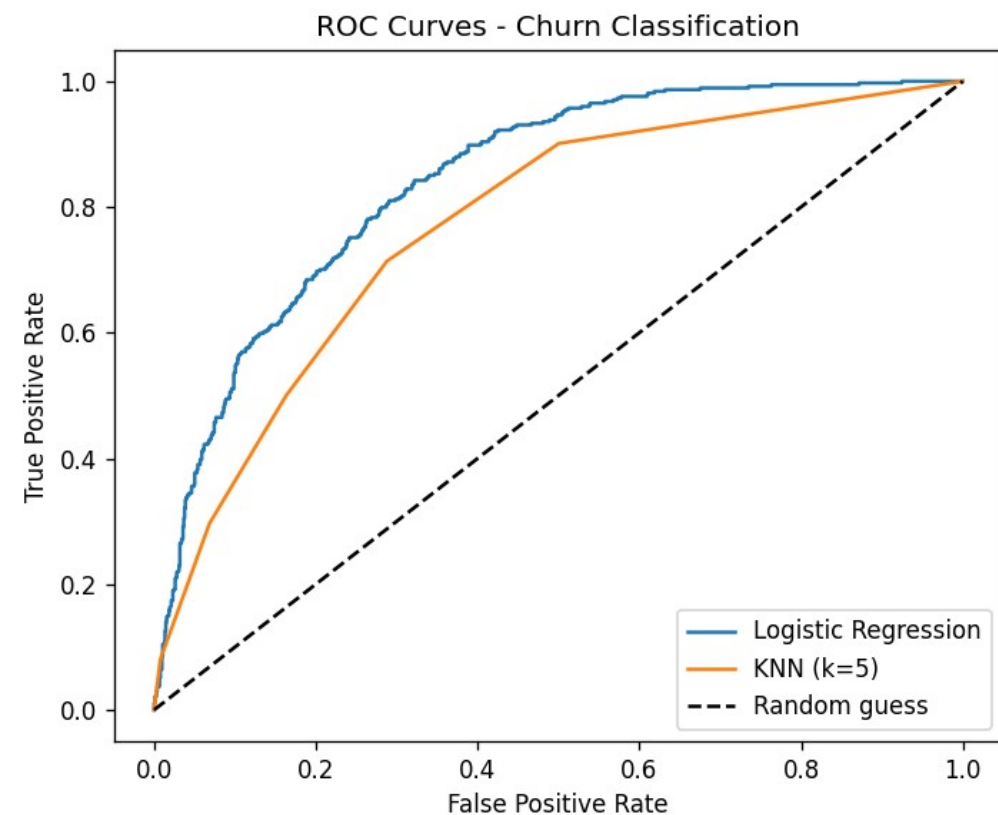
Models used: Classification → Logistic Regression vs K-Nearest Neighbours • Regression → Linear Regression vs KNN Regressor

● Results — Classification (Churn)

Metric	Logistic Regression	KNN (k = 5)
Accuracy	0.807	0.747
Precision	0.658	0.525
Recall	0.567	0.500
F1-Score	0.609	0.512
ROC-AUC	0.842	0.772

WINNER: Logistic Regression

Better on every metric — higher accuracy and far stronger ROC-AUC.

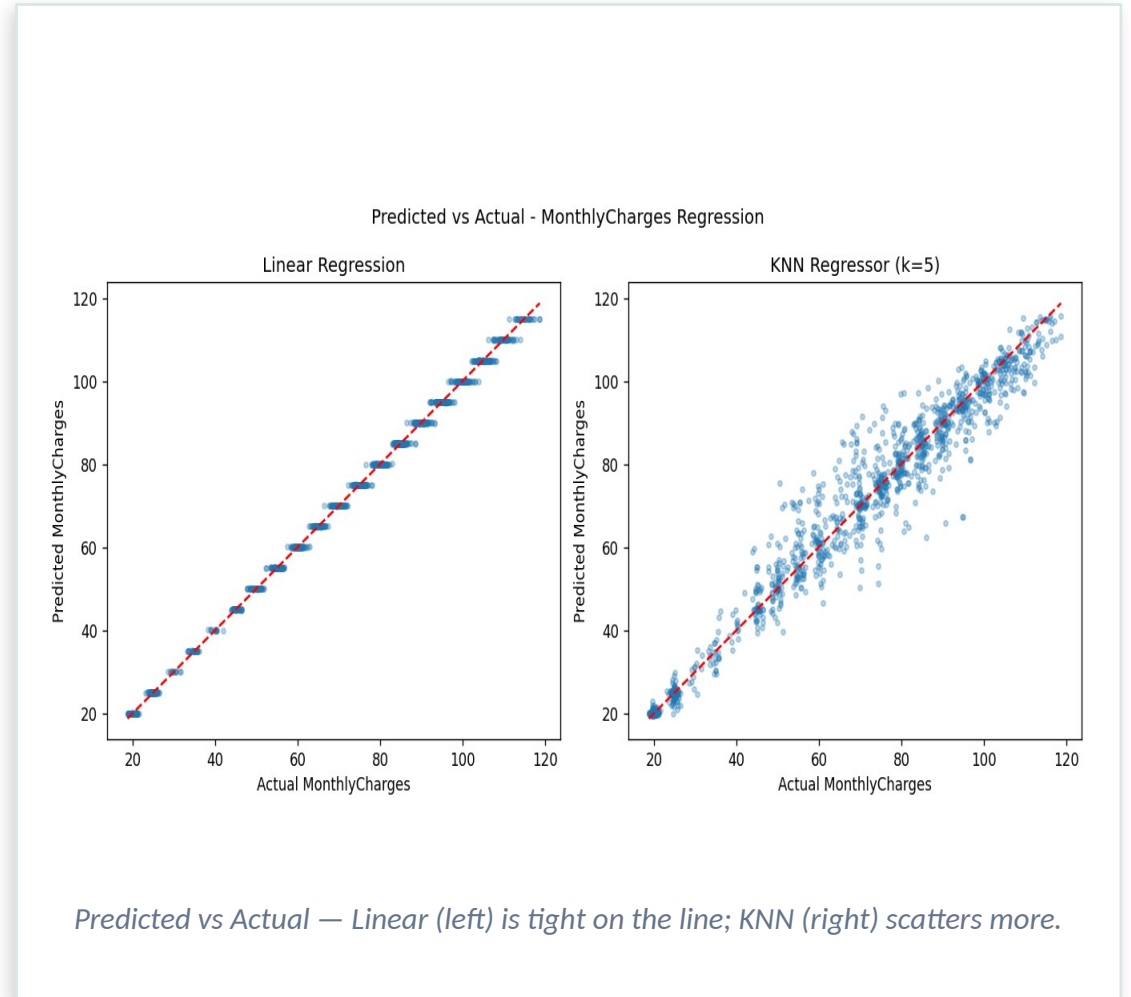


ROC curve: the higher (more top-left) the curve, the better the model.

● Results — Regression (MonthlyCharges)

Metric	Linear Regression	KNN Regressor
MAE (lower better)	0.79	3.46
MSE (lower better)	1.11	29.19
RMSE (lower better)	1.05	5.41
R ² (higher better)	0.999	0.968
WINNER: Linear Regression Average error under \$1 and R ² ≈ 0.999 — near-perfect fit.		

Points hugging the red line = accurate predictions.



Comparative Analysis

Why Logistic Regression won

After one-hot encoding there are ~30 feature columns. KNN measures distance between customers, and in high dimensions distances blur — the "curse of dimensionality". Logistic Regression learns one weight per feature, so many columns don't confuse it.

Why Linear Regression won

Monthly charge is essentially the sum of per-service fees — a genuinely linear, additive relationship. That is exactly what Linear Regression models, so it fits almost perfectly. KNN only averages similar customers, which is fuzzier.

Key takeaway

On this dataset the simpler linear models beat KNN in BOTH tasks — because the underlying patterns are linear, and KNN struggles once there are many features. The right model depends on the shape of the data, not on complexity.

Trained on 5,634 customers • evaluated on 1,409 unseen customers • 4 models • single workflow